Alberto Andrés Valdés González.
Degree: Mathematical Engineer.
Work position: ML-Engineer.
Mail: anvaldes@uc.cl/alberto.valdes.gonzalez.96@gmail.com
Location: Santiago, Chile.

Bootstrap Confidence Intervals for Evaluation Metrics with Repeated Runs

---

Problem: a model is evaluated on N examples, and since its output is stochastic, each example is evaluated R times. We want a **confidence interval** for the mean score. The $N \cdot R$ values are **not independent**: runs of the same example are correlated, so treating them as $N \cdot R$ samples gives intervals that are too narrow. **Idea:** treat each example as a **cluster**, average its runs, and bootstrap over **examples**, not over individual runs.

Notation:

$$X_{(i)}^{(j)} = \text{value of the evaluation for run } j \text{ of example } i, \qquad j \in \{1, \dots, R\}, \quad i \in \{1, \dots, N\}$$

Mean per example and **global point estimate**:

$$\bar{X}_{(i)} = \frac{1}{R} \sum_{j=1}^{R} X_{(i)}^{(j)}, \qquad \hat{\theta} = \frac{1}{N} \sum_{i=1}^{N} \bar{X}_{(i)} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{R} \sum_{j=1}^{R} X_{(i)}^{(j)} = \frac{1}{N \cdot R} \sum_{i=1}^{N} \sum_{j=1}^{R} X_{(i)}^{(j)}$$

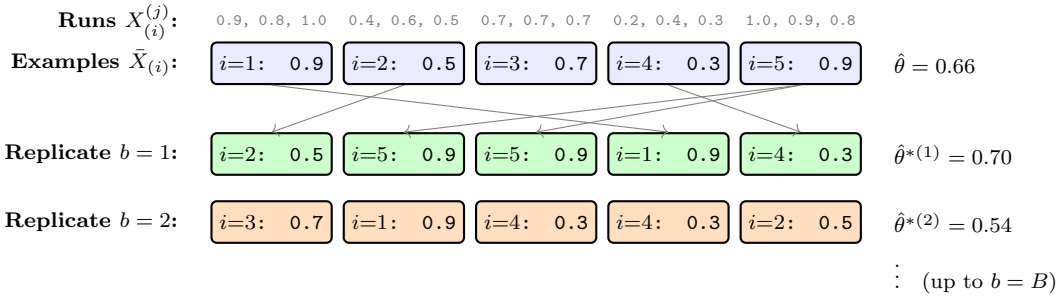
Resampling with bootstrap (for each $b \in \{1, \dots, B\}$):

(1) **Sample indices:** draw N indices with replacement: $i_{(1)}^{(b)}, \dots, i_{(N)}^{(b)} \overset{\text{iid}}{\sim} \mathcal{U}\{1, \dots, N\}$.

(2) **Replicate:** average the per-example means of the sampled examples: $\hat{\theta}^{*(b)} = \frac{1}{N} \sum_{n=1}^{N} \bar{X}_{\left(i_{(n)}^{(b)}\right)}$.

An example can appear several times or not at all in one replicate, but its R runs **always travel together**.

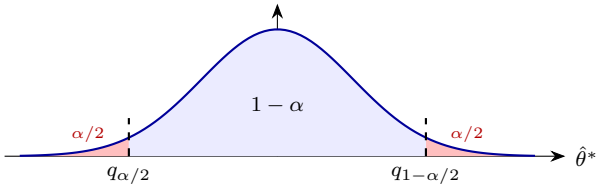
Example: $N = 5$ examples, $R = 3$ runs each.



In replicate $b = 1$, example 5 was drawn twice and example 3 was left out. Repeating this $B = 10{,}000$ times gives the bootstrap distribution of $\hat{\theta}^*$.

Confidence interval of level $(1 - \alpha)$**:** sort the replicates, $\hat{\theta}^{*[1]} \leq \hat{\theta}^{*[2]} \leq \cdots \leq \hat{\theta}^{*[B]}$, and take the empirical quantiles:

$$q_p = \hat{\theta}^{*[\lfloor p \cdot B \rfloor]}, \qquad \boxed{\mathrm{CI}_{(1-\alpha)} = \left[\, q_{\alpha/2},\ q_{1-\alpha/2} \,\right]}$$



The **percentile** interval is **robust to the distribution**: it reads the quantiles directly from the replicates and does not assume normality. In the example, with $\alpha = 0.05$: $\mathrm{CI}_{95\%} = [0.46, 0.86]$ (wide, since $N = 5$ is tiny).

Alternatives: basic (pivotal) $\left[\, 2\hat{\theta} - q_{1-\alpha/2},\ 2\hat{\theta} - q_{\alpha/2} \,\right]$, and **normal** $\hat{\theta} \pm z_{1-\alpha/2}\, \widehat{se}_B$, with

$$\widehat{se}_B = \sqrt{\frac{1}{B-1} \sum_{b=1}^{B} \left(\hat{\theta}^{*(b)} - \bar{\theta}^* \right)^2}, \qquad \bar{\theta}^* = \frac{1}{B} \sum_{b=1}^{B} \hat{\theta}^{*(b)}.$$

Remarks:

- **Resample only the examples:** each $\bar{X}_{(i)}$ already carries the run-to-run noise, so the interval captures both sources of variability (between and within examples). Also resampling the runs (two-level bootstrap) tends to **overestimate** the variance.
- **Sanity check:** as $B \to \infty$, $\widehat{se}_B^{\,2} \to \frac{1}{N^2} \sum_{i=1}^{N} \left(\bar{X}_{(i)} - \hat{\theta} \right)^2$. In the example: $\widehat{se}_B \approx 0.104$ in both cases.
- **Choice of** B**:** $B \approx 10^3$ is enough for $\widehat{se}_B$; use $B \geq 10^4$ for percentile intervals at 95% or more.

Implementation (`X` with shape `(N, R)`):

```
Xbar_i = X.mean(axis=1)                          # (N,)  mean per example
idx    = rng.integers(0, N, size=(B, N))         # (B, N) resampled indices
theta  = Xbar_i[idx].mean(axis=1)                # (B,)  bootstrap replicates
ci     = np.quantile(theta, [alpha/2, 1 - alpha/2])  # percentile CI
```